Stationary

{max2} is the best. ***
{max1} is better than {min2}, {min1}, and {mid}. ***
{min2} is the worst.
One button is the best.
Explore all buttons, then stick with the one that gave the most points.
No hint (model vs. people)

0 5 10 15
Regret reduction (%)

Drifting

{max2} begins as the best, but the button values change over trials. **
{min2} and {min1} begin as the worst buttons.
The button values drift by a small number at each trial.
The button values change gradually.
Even after finding a good button, continue trying the others to detect changes.
No hint (model vs. people)

−10 0 10 20
Regret reduction (%)

Time Delayed

{max2} is the best, even if it doesn't seem like it at first. ***
{max2} is the best. ***
Choose a button for a few trials in a row to observe its true value.
The points correspond to the choice you made 3 trials ago.
There is a time delay between your choice and the points you receive.
No hint (model vs. people)

0 20 40
Regret reduction (%)

Two Context

Find a few good buttons and alternate between them to maximize your points. ***
{max2} begins as the best, but the points decrease over time.
Do not stick with any one button for too long.
Each button value will decrease if you keep choosing it, but it will increase over time if not.
The points you get on the first pick of a button are the most representative of the button value. **
No hint (model vs. people)

−5 0 5 10
Regret reduction (%)

Three Context

When you see a square, {max2} is the best. When you see a circle, {min2} is the best. ***
The button values change for square and circle.
Each shape corresponds to a different set of button values.
Pay attention to the shapes.
Explore the buttons to find the best one for each shape.
No hint (model vs. people)

0 20
Regret reduction (%)

Bernoulli

When you see a square, choose G. When you see a circle, choose A. When you see a triangle, explore all buttons to find the best. ***
Pay attention to the shapes.
Explore the buttons to find the best one for each shape.
The button values change under each shape.
When you see a square or circle, the buttons are sorted in different directions.
No hint (model vs. people)

0 10 20
Regret reduction (%)

Linear Contextual

{max2} is most likely to return 100 points. ***
Explore all buttons, then stick to the one that most often returned points.
The probability for returning points is fixed for each button.
Choose a button multiple times to learn its probability of returning points.
There is a chance of returning 0 points even if the button is good. *
No hint (model vs. people)

0 20
Regret reduction (%)

Bernoulli

If the third number under the button is 1, you will get the most points. ***
Choose the button with a 1 as its third number. ***
One of the three numbers under the buttons always leads to more points.
The first number has a negative impact on the points. Only choose buttons with a 0 as the first number.
The three numbers under each button combine to calculate the total points.
No hint (model vs. people)

0 20 40
Regret reduction (%)

Linear Contextual

Human Bayesian Centaur

Human no-hint ± 95% CI